Lecture 7

Array Subsystems

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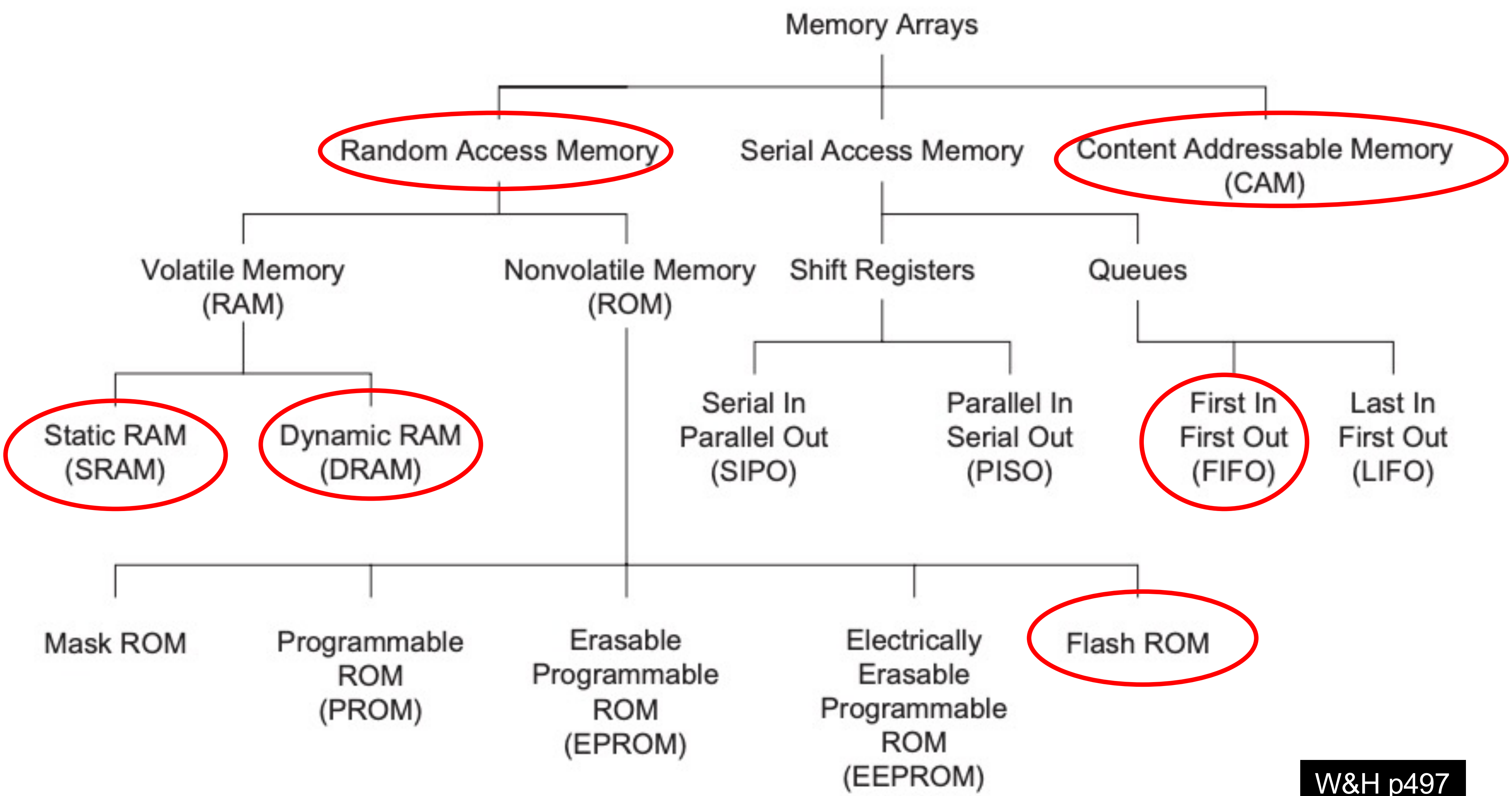
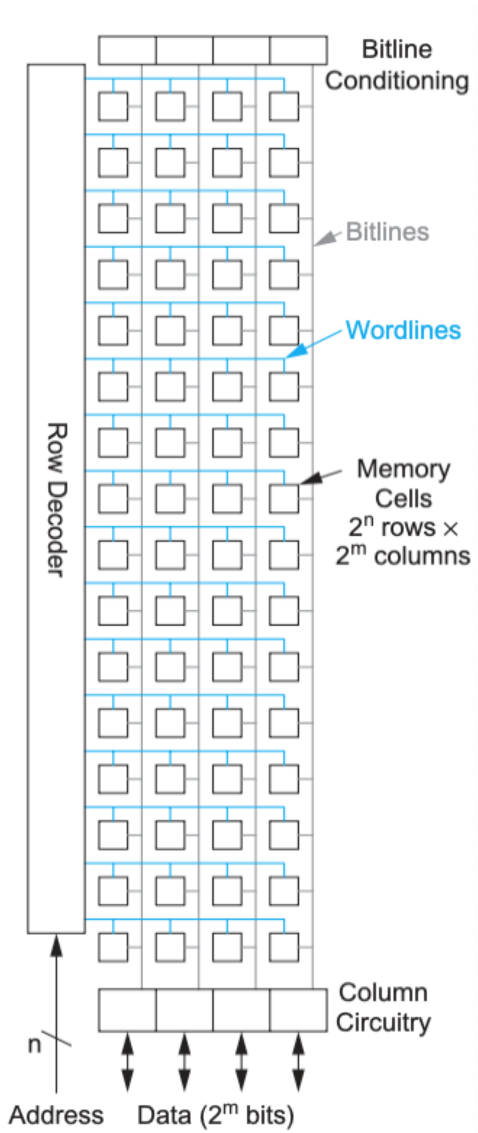
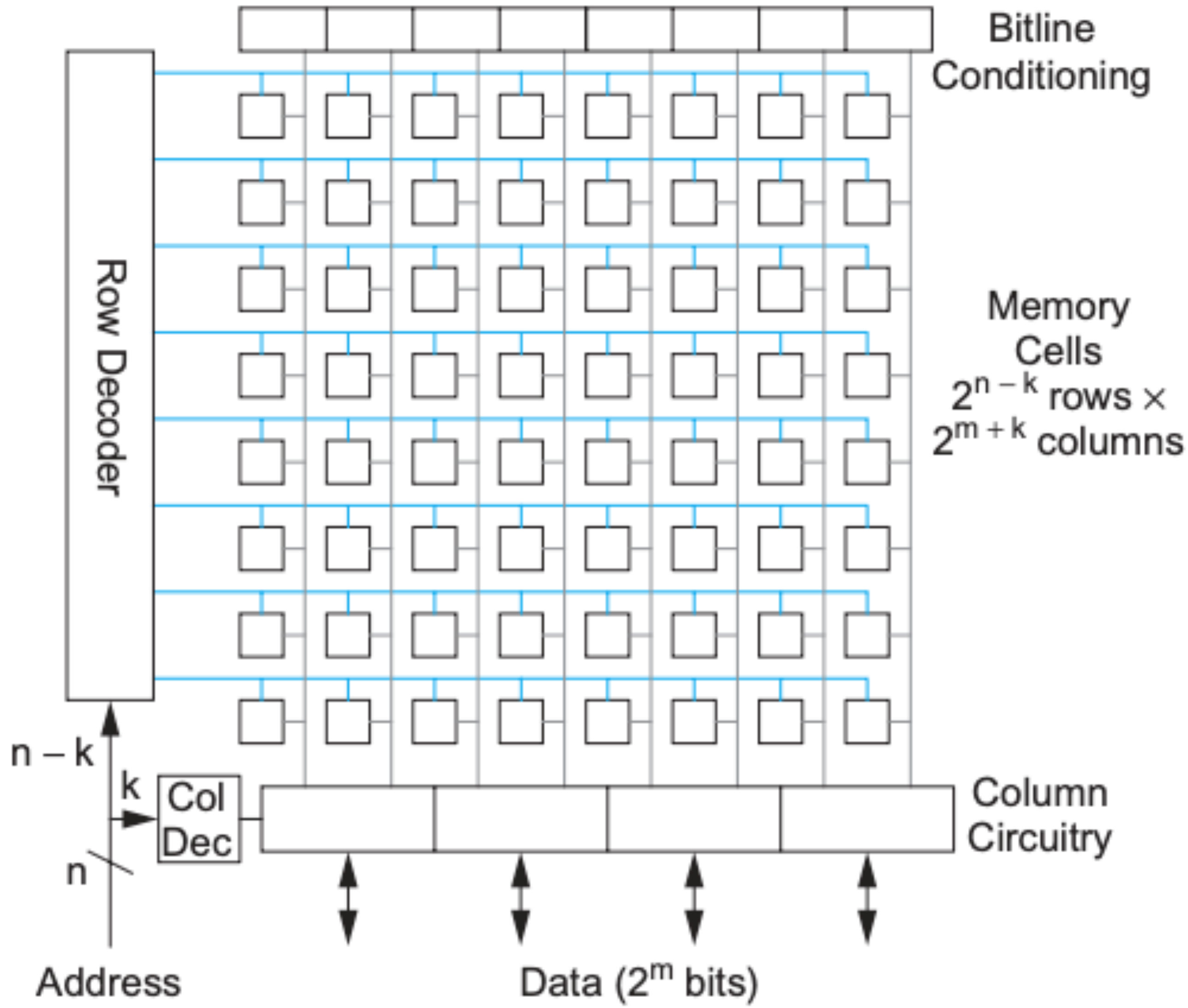
Categories of Memory Arrays



W&H p497

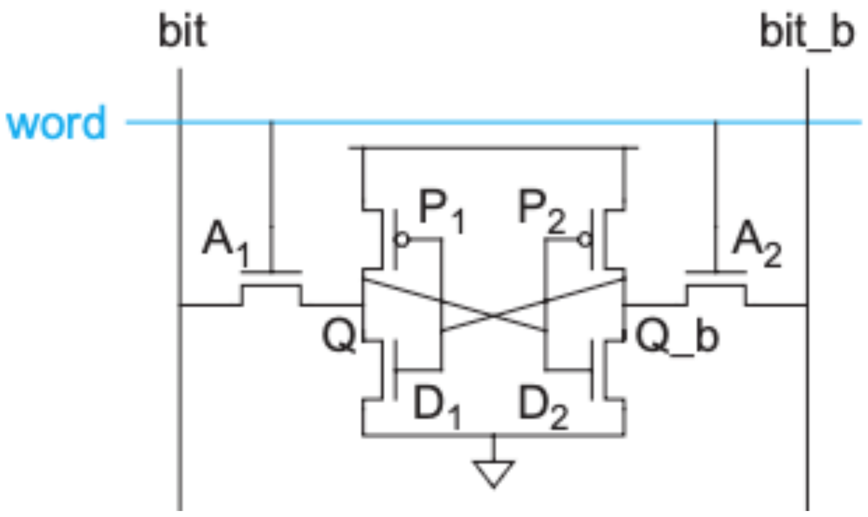
Memory Array Architecture

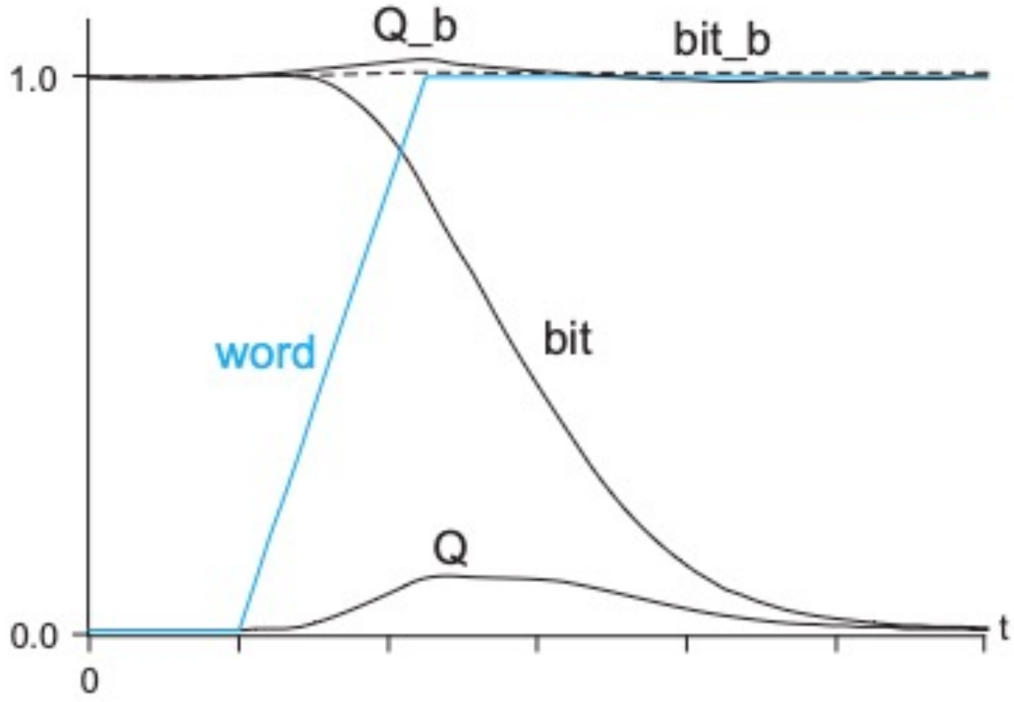




W&H p499-500

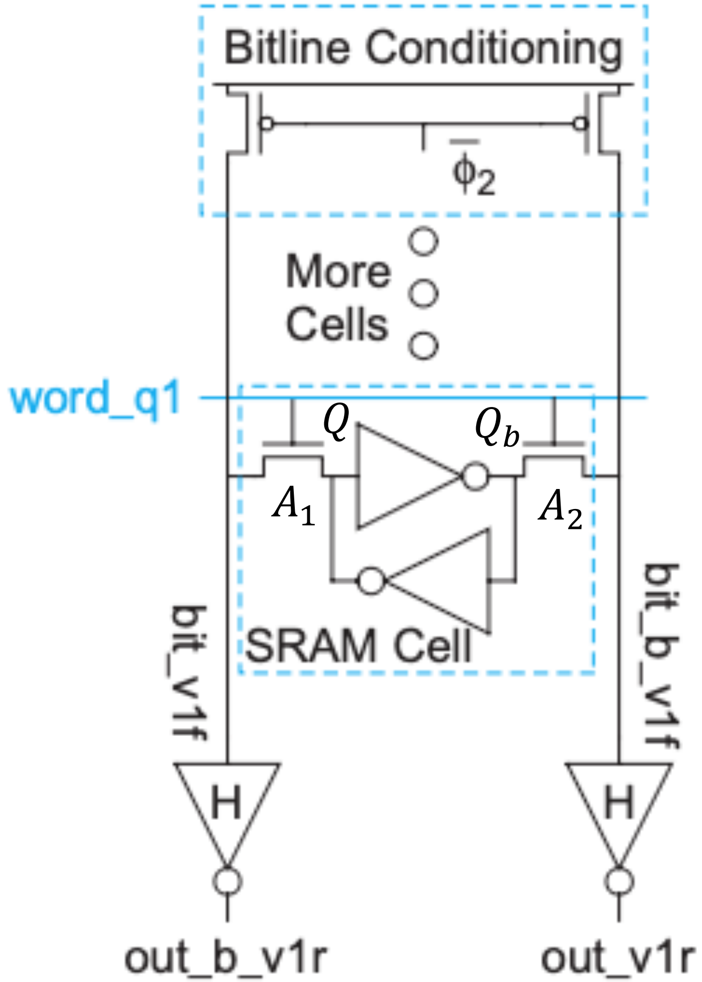
Read operation for SRAM cell

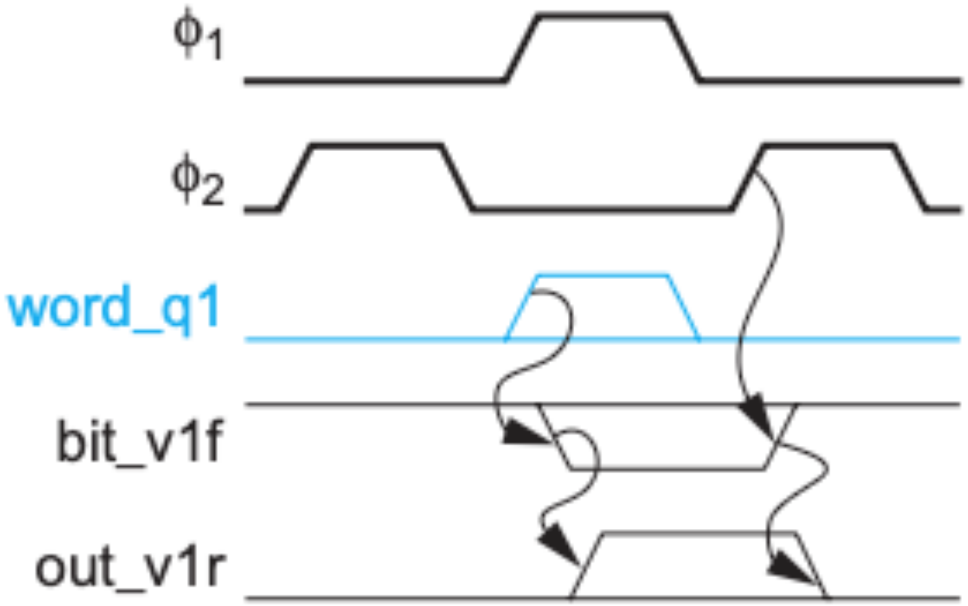




W&H p500

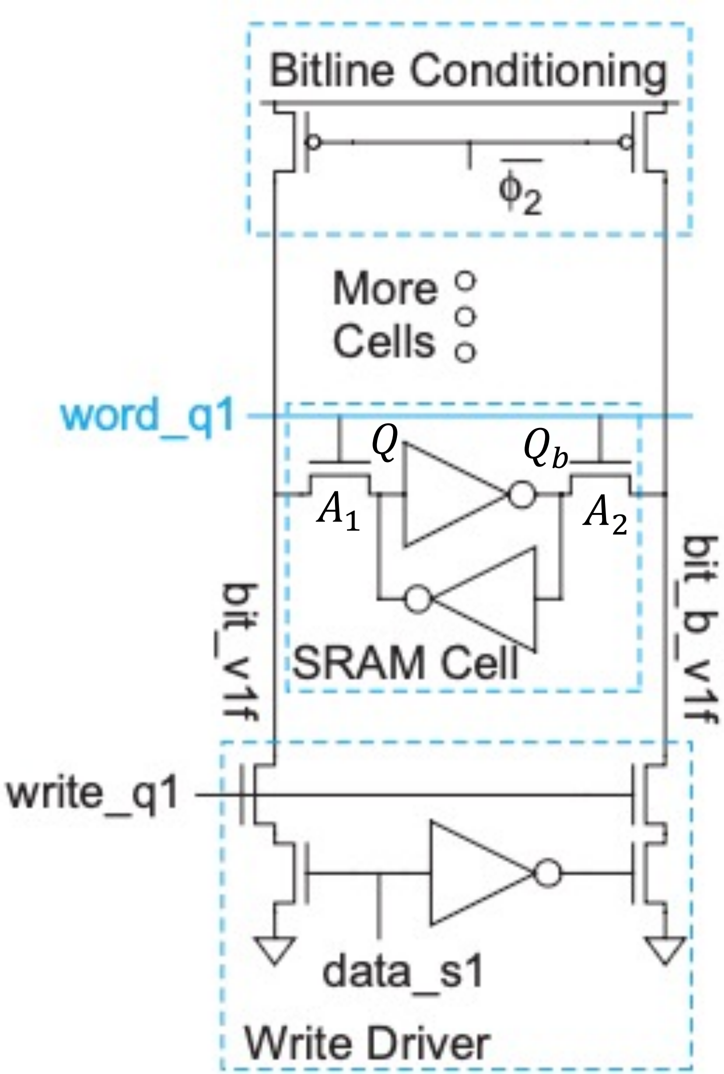
Column Read Operation

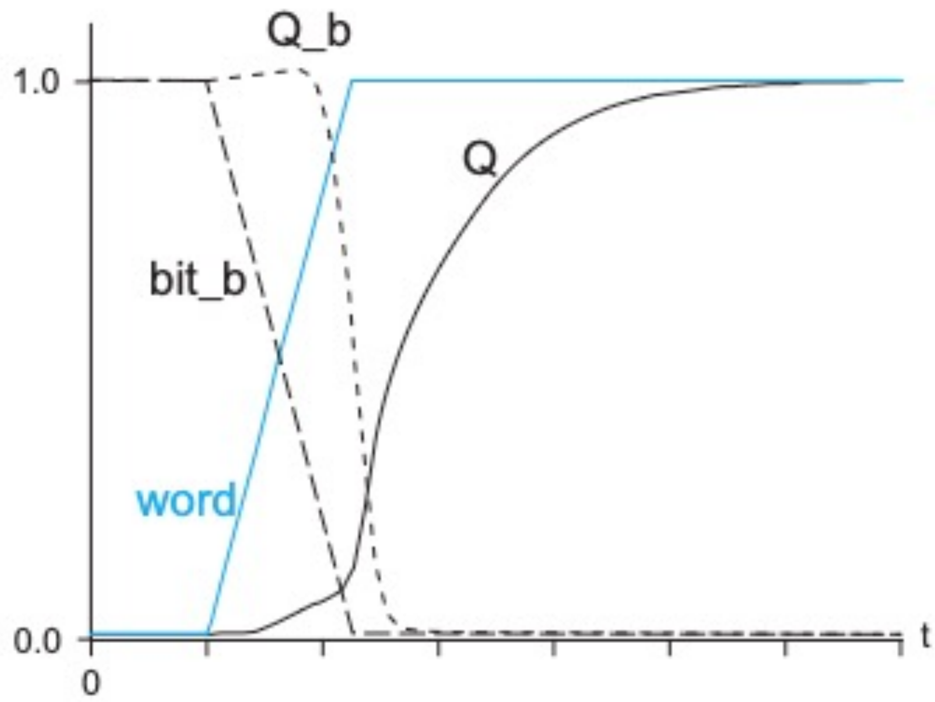


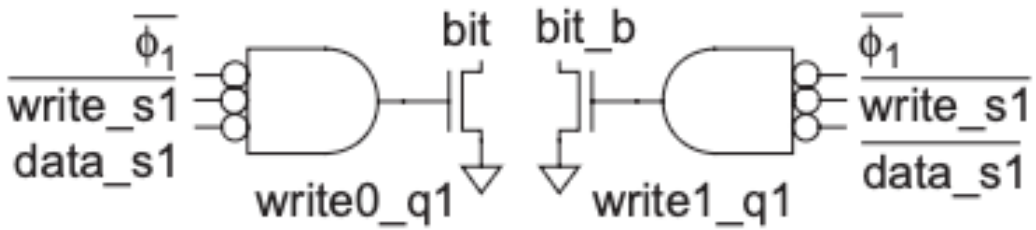


W&H p501

Write Operation

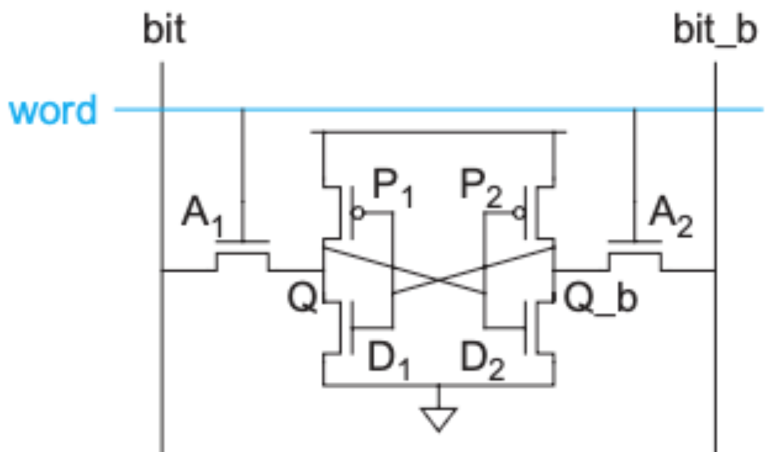


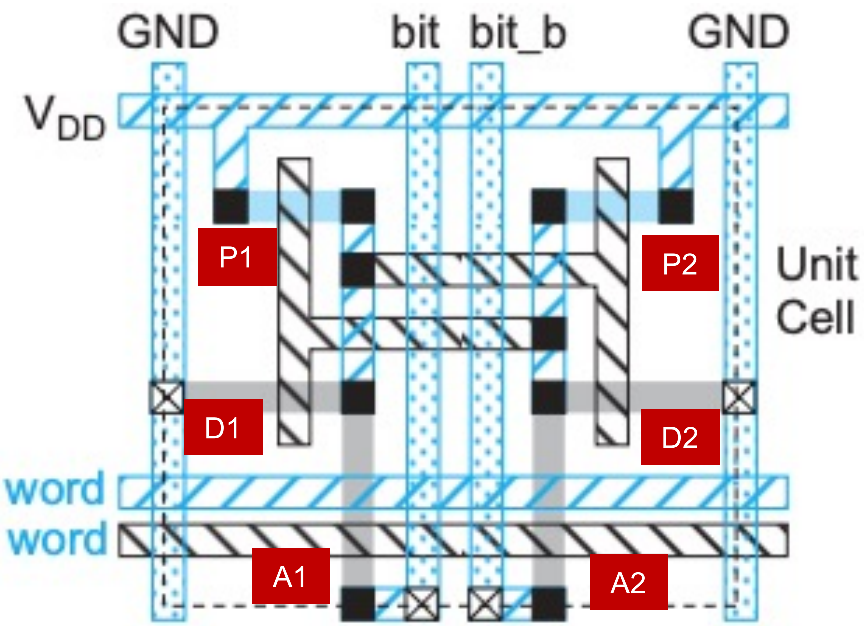


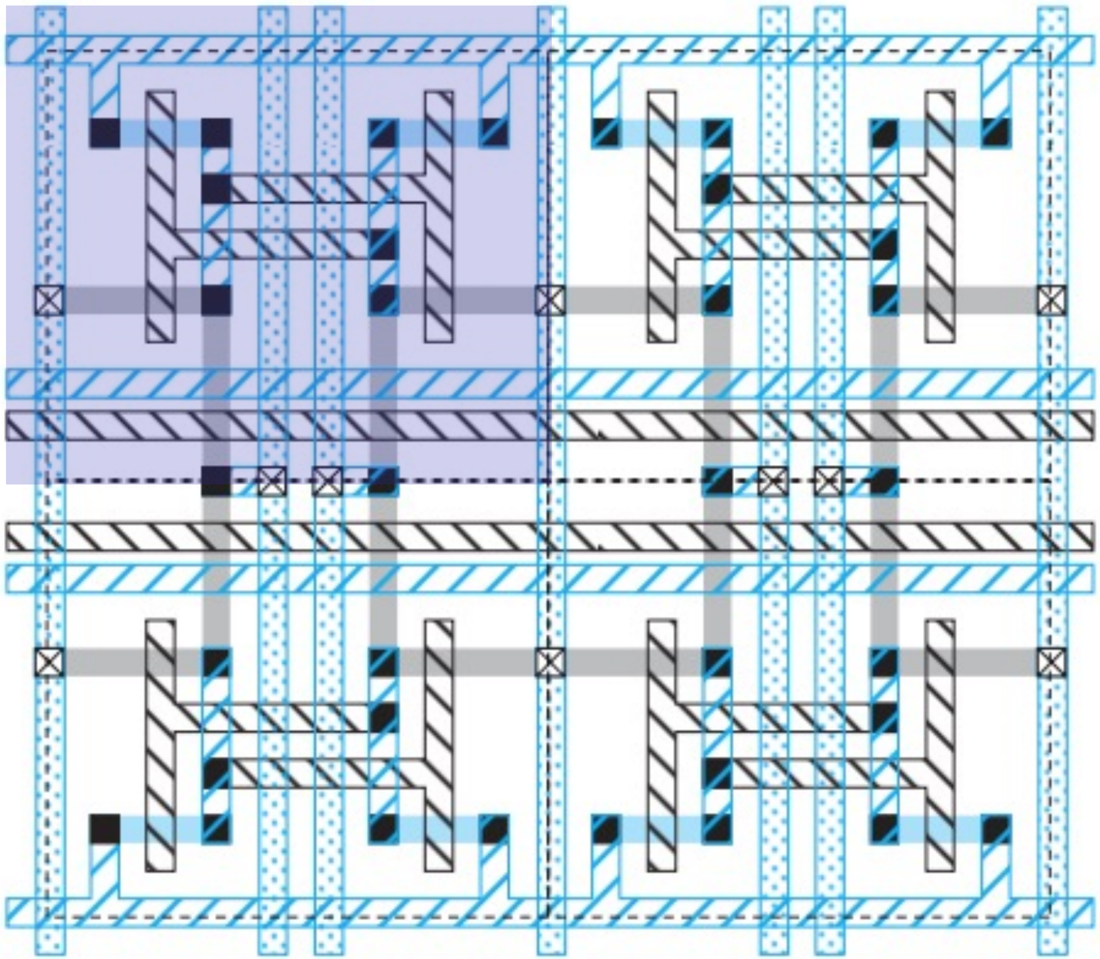


W&H p501

6T SRAM cell Layout (stick diagram)

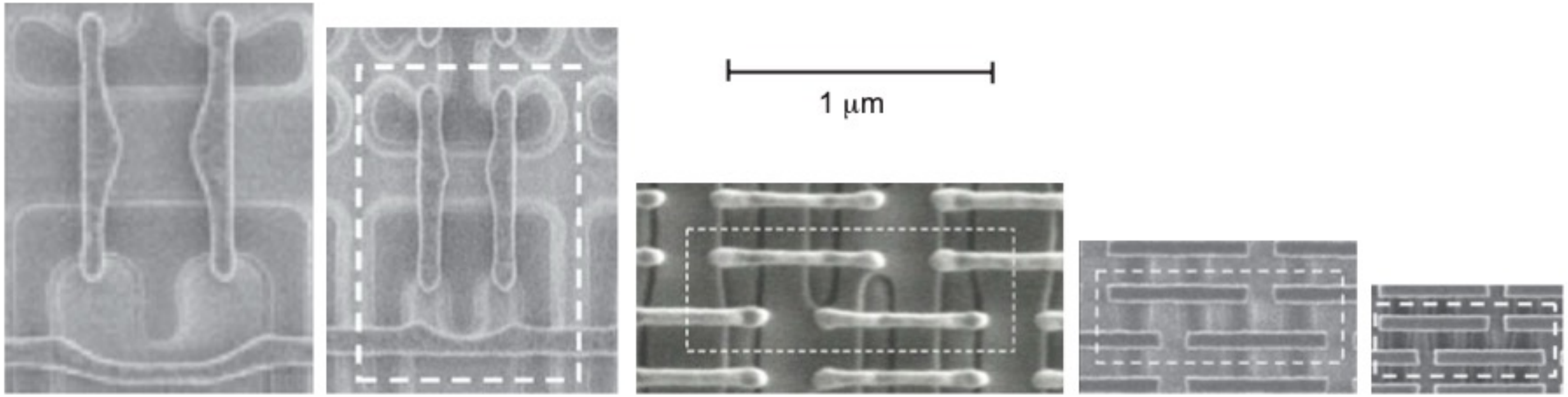




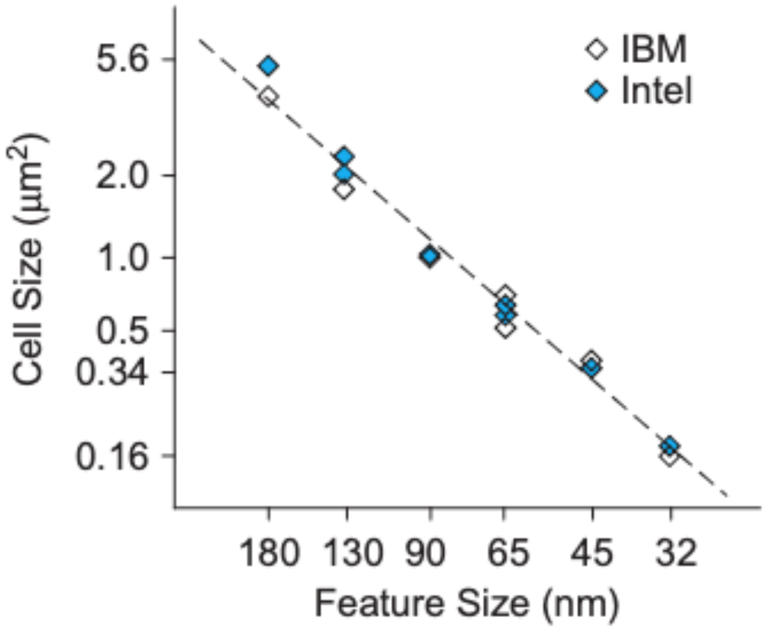


W&H p504

SRAM Scaling

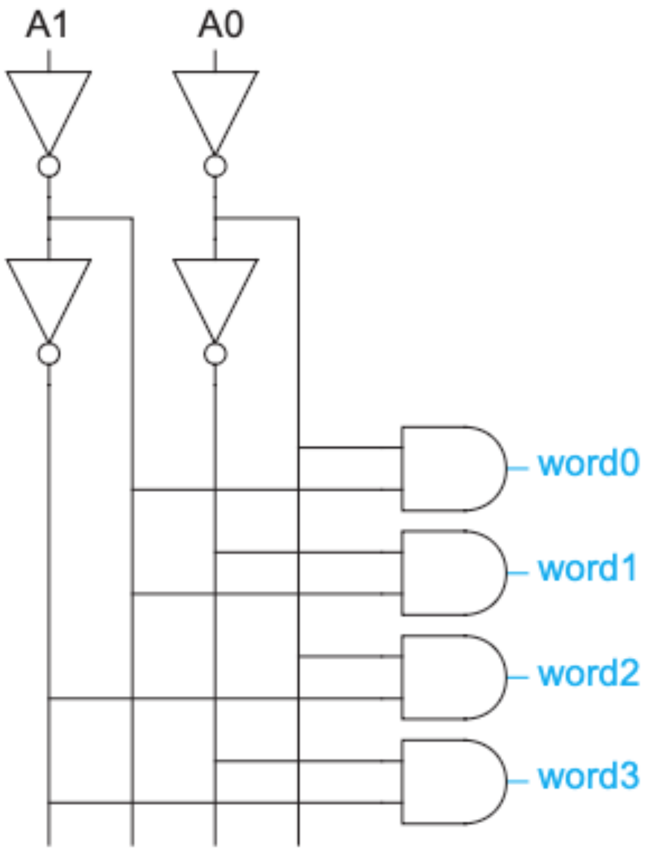


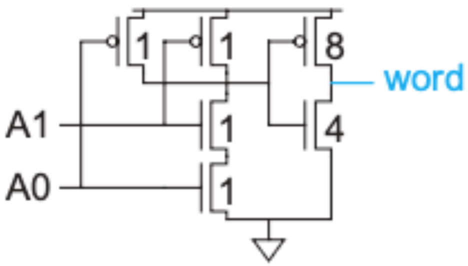
130 nm [Tyagi00] 90 nm [Thompson02] 65 nm [Bai04] 45 nm [Mistry07] 32 nm [Natarajan08]

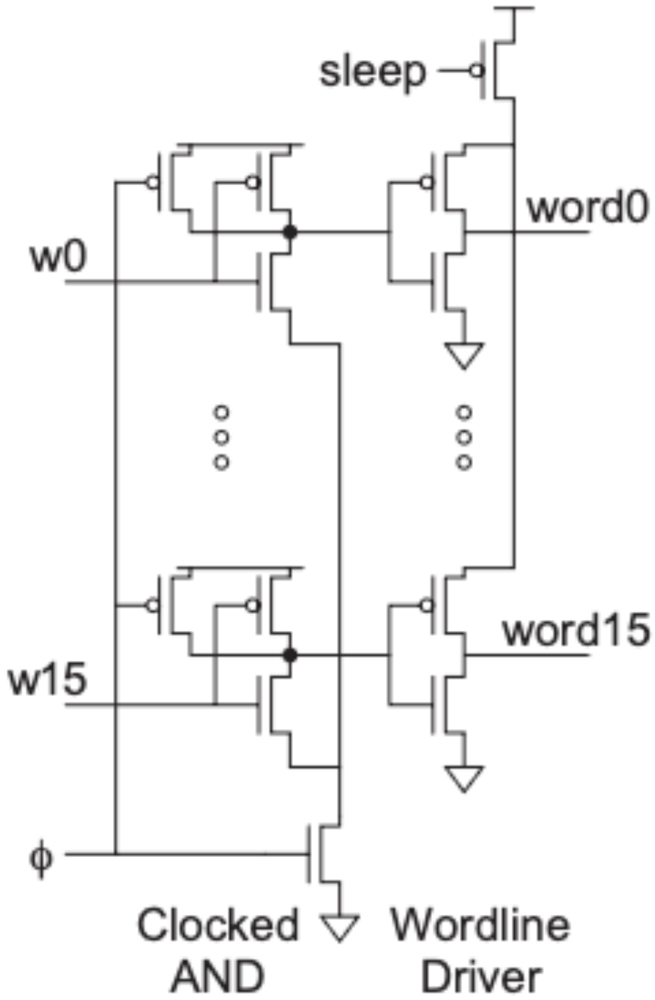


W&H p505

Row Decoder Circuits



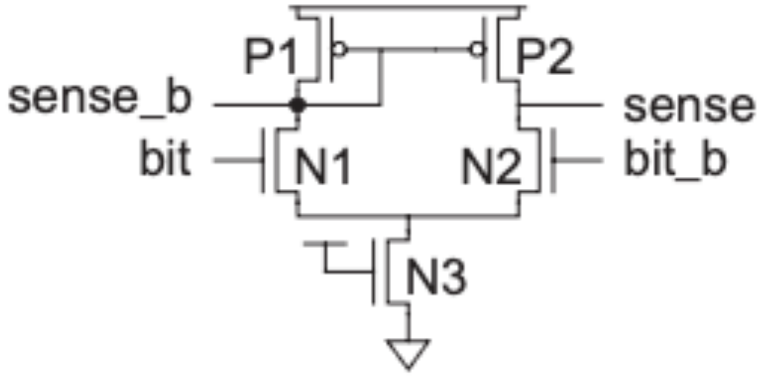




W&H p506-507

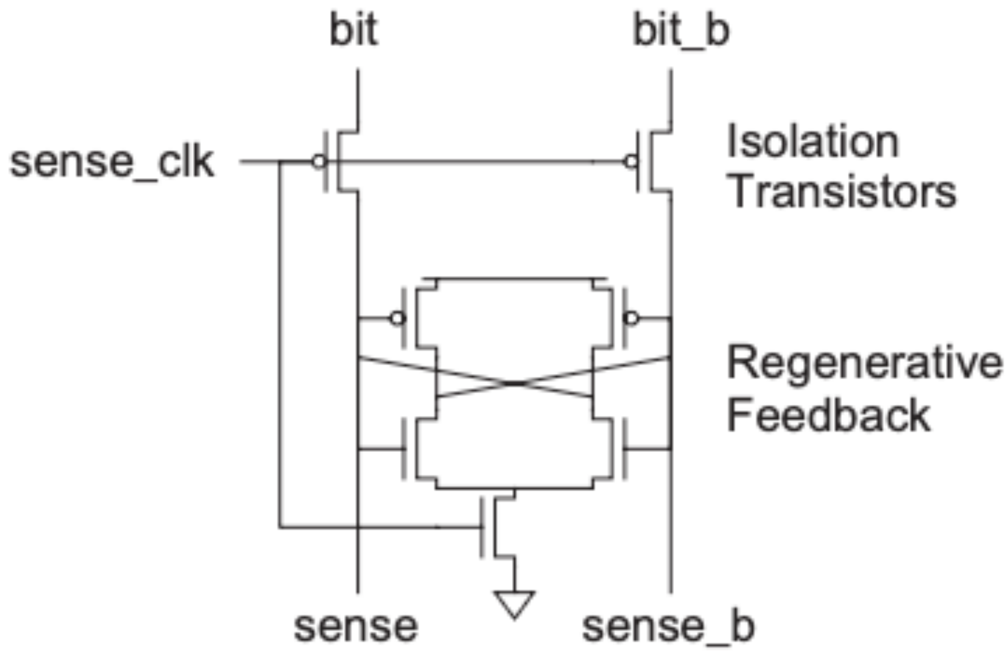
Sense Amplifier

Differntial Sense Amplifier



(a)

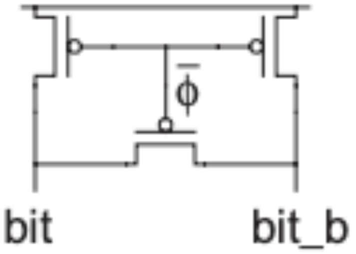
Clocked Sense Amplifier



(b)

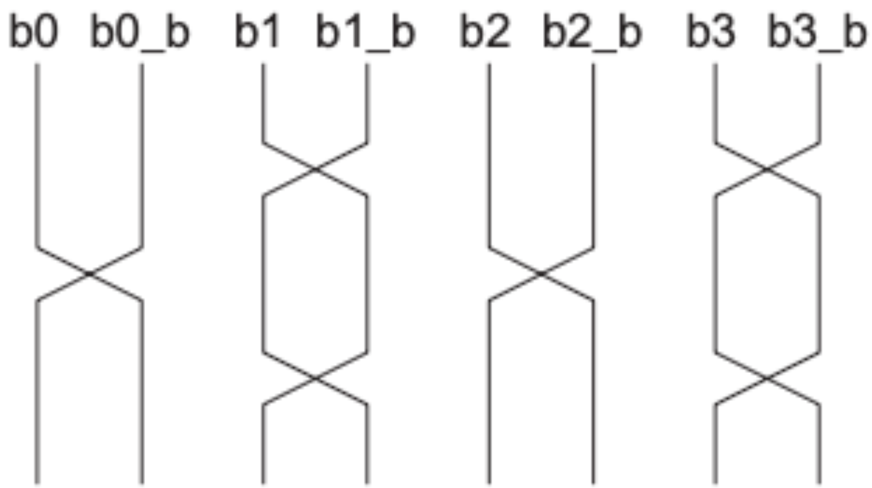
W&H p511

Differential Noise Reduction



(a)

Bitlines equalization

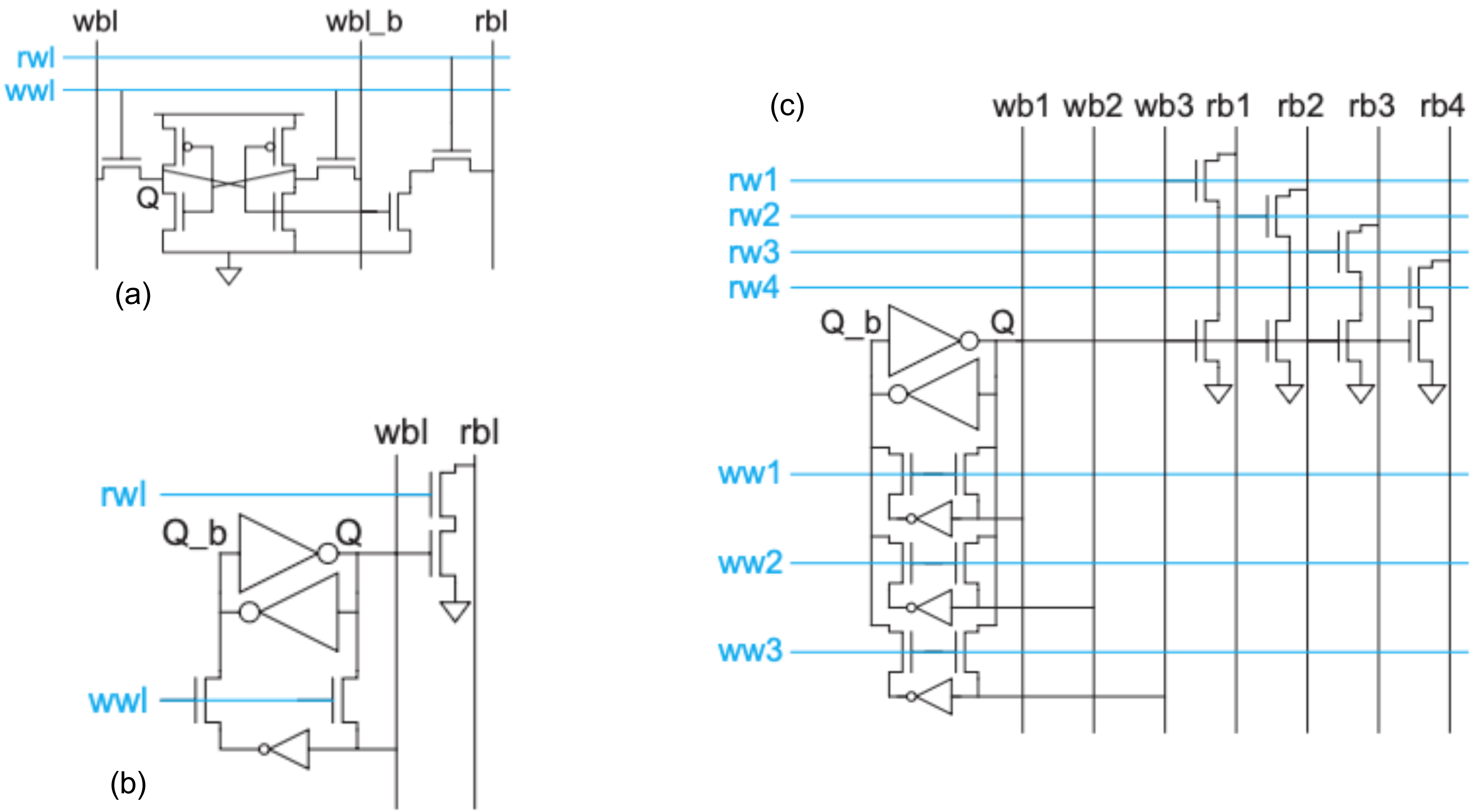


(b)

Bitlines twisting

W&H p513

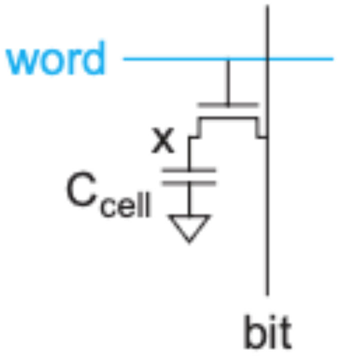
Multiport SRAM

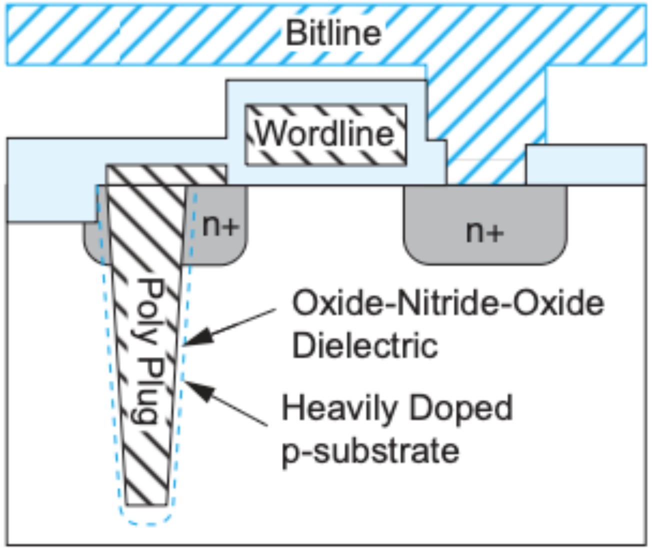


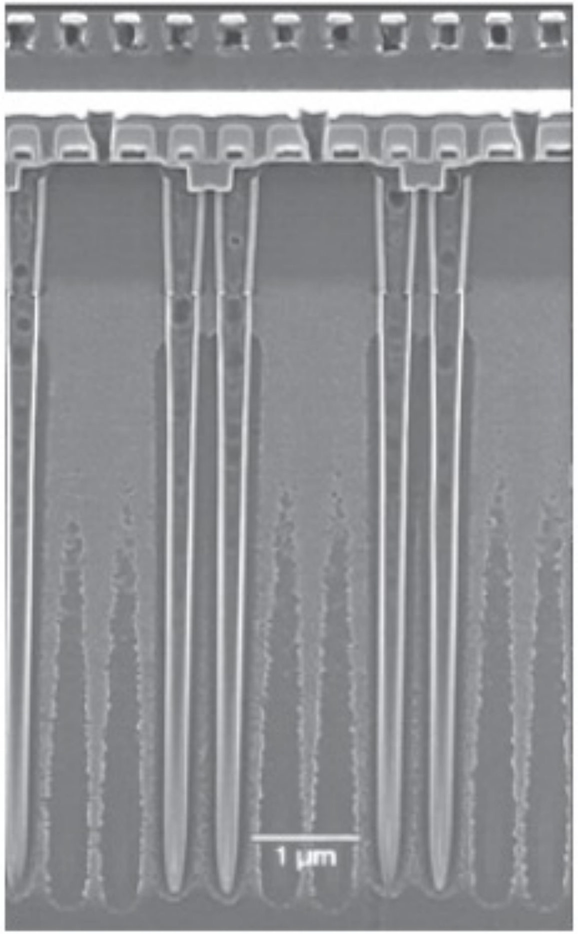
W&H p514-515

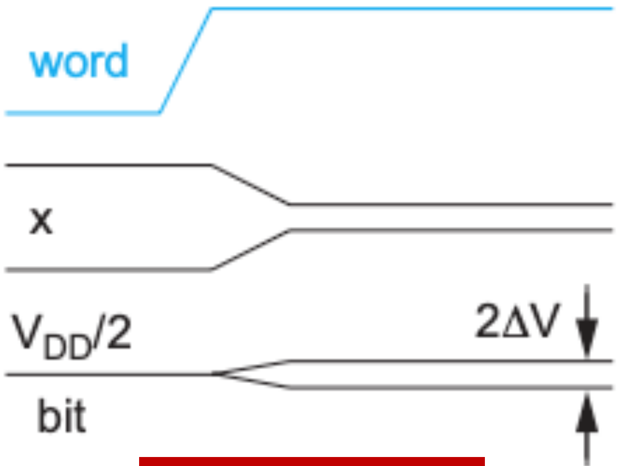
DRAM cell

Trench capacitor









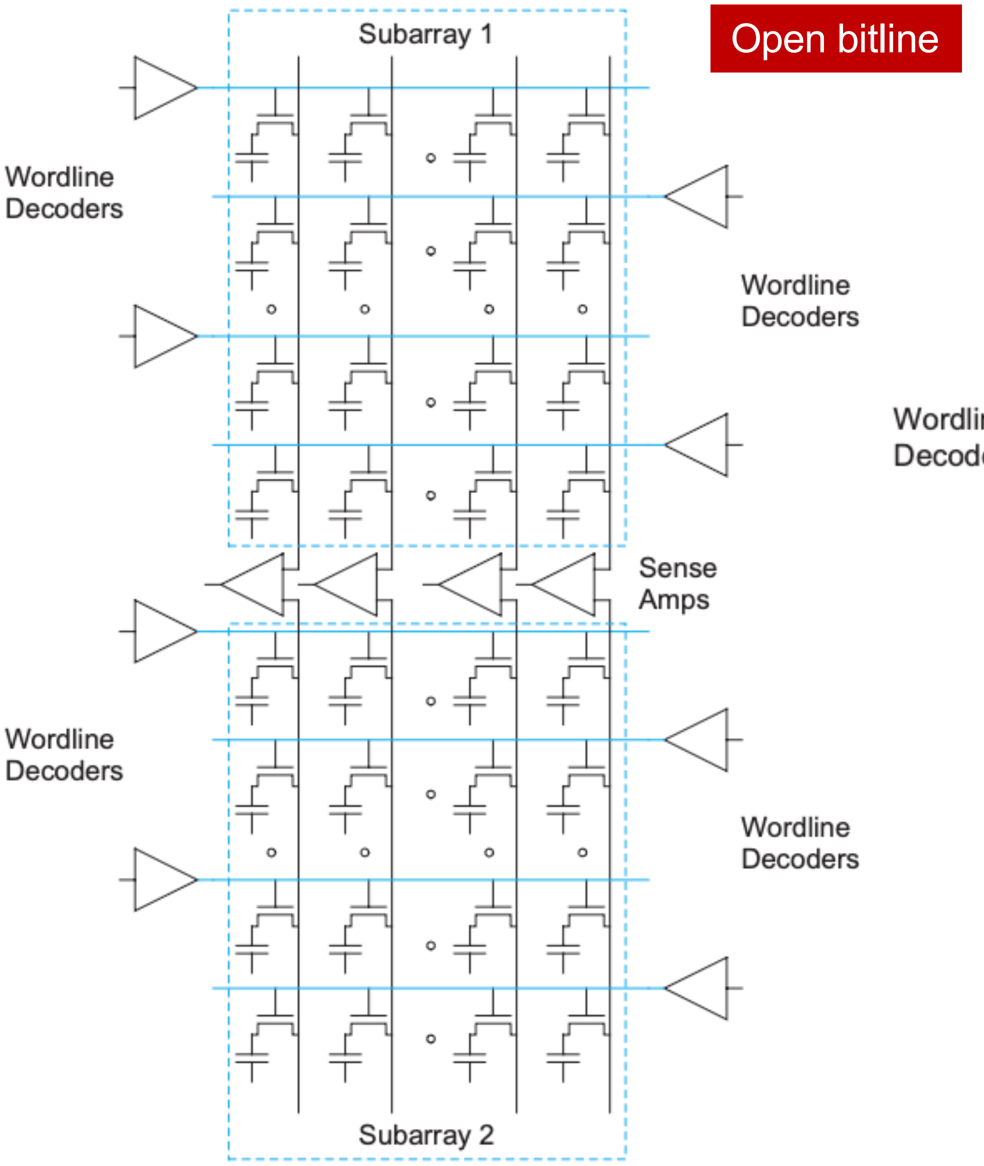
VDD/2 precharge

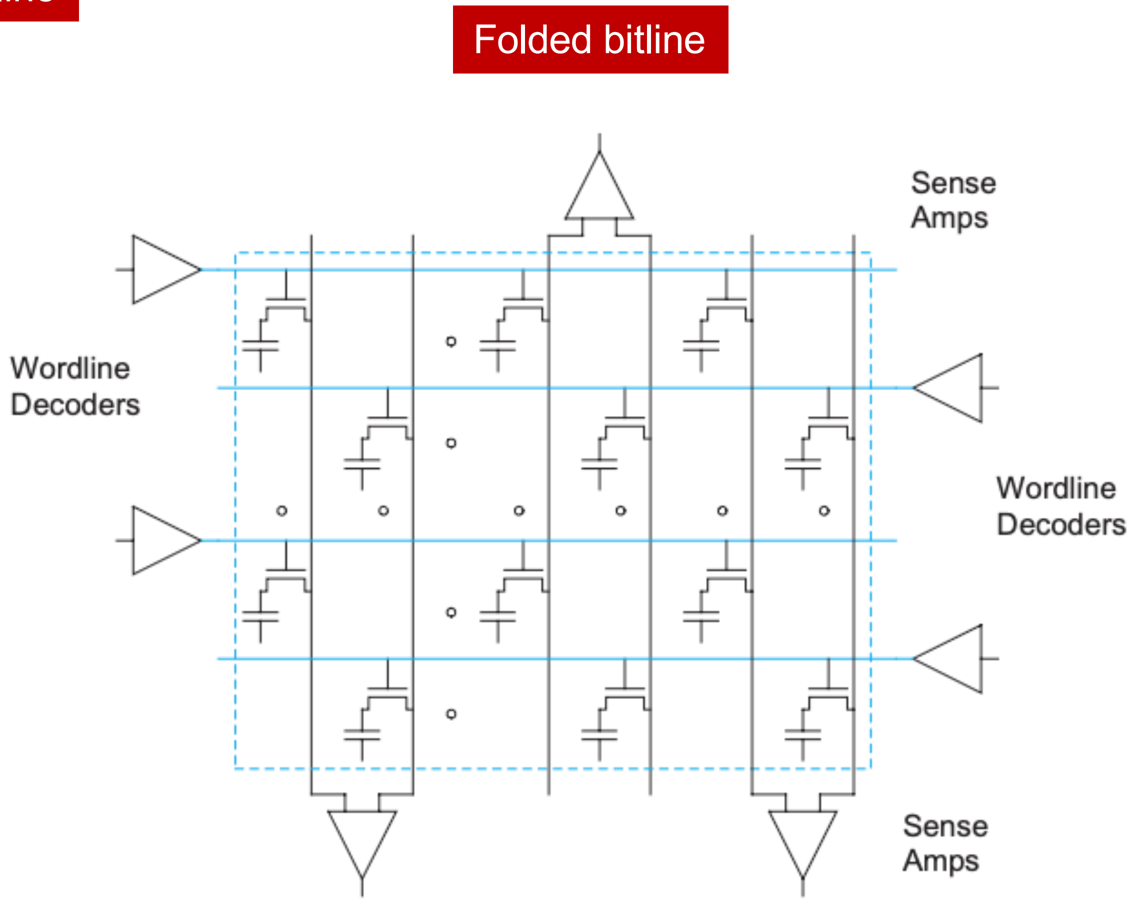
$$\Delta V = \frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell}+C_{bit}}$$

Charge sharing

W&H p522-523

DRAM array

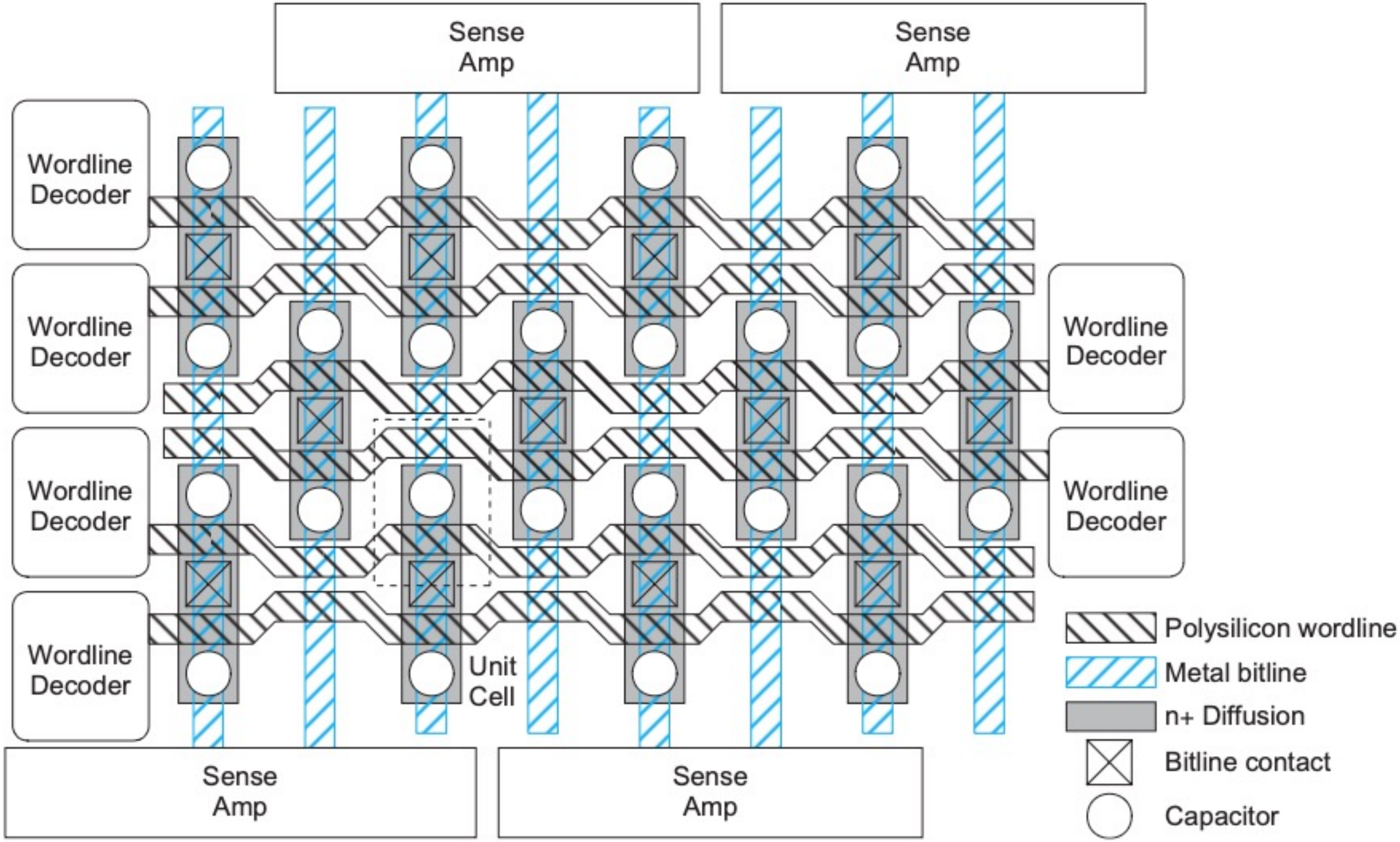




- Neighbouring bitline provide reference.
- Reduce noise sensitibity

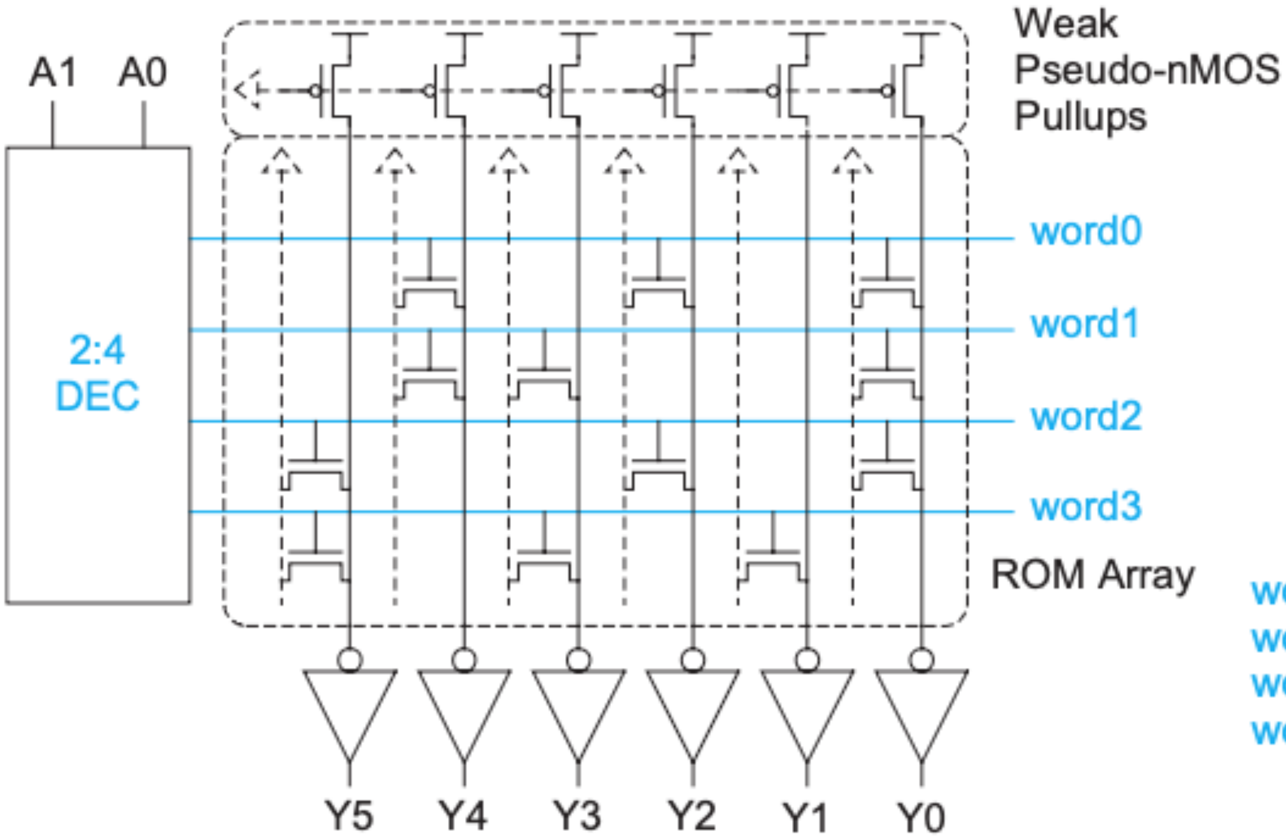
W&H p524

Folded Bitline Layout



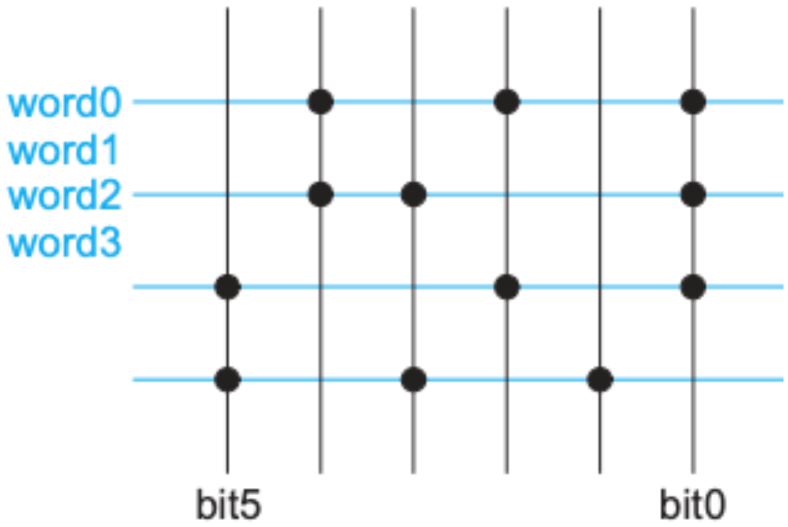
W&H p525

Pseudo nMOS ROM Circuit



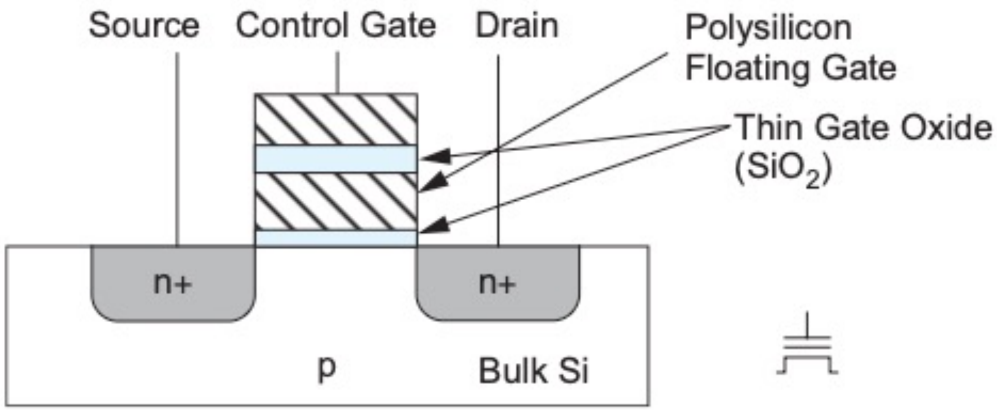
4-word by 6-bit ROM

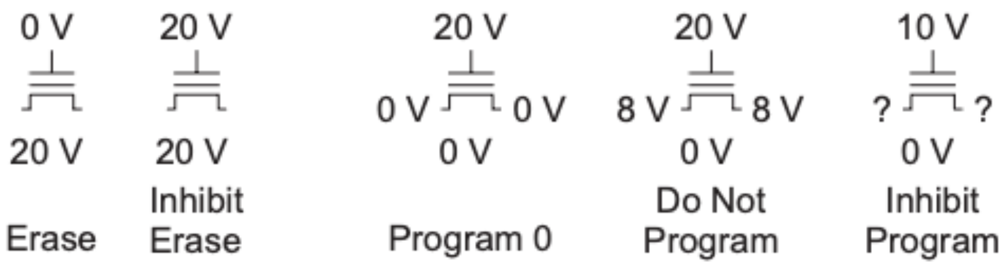
```
word0: 010101
word1: 011001
word2: 100101
word3: 101010
```



W&H p527

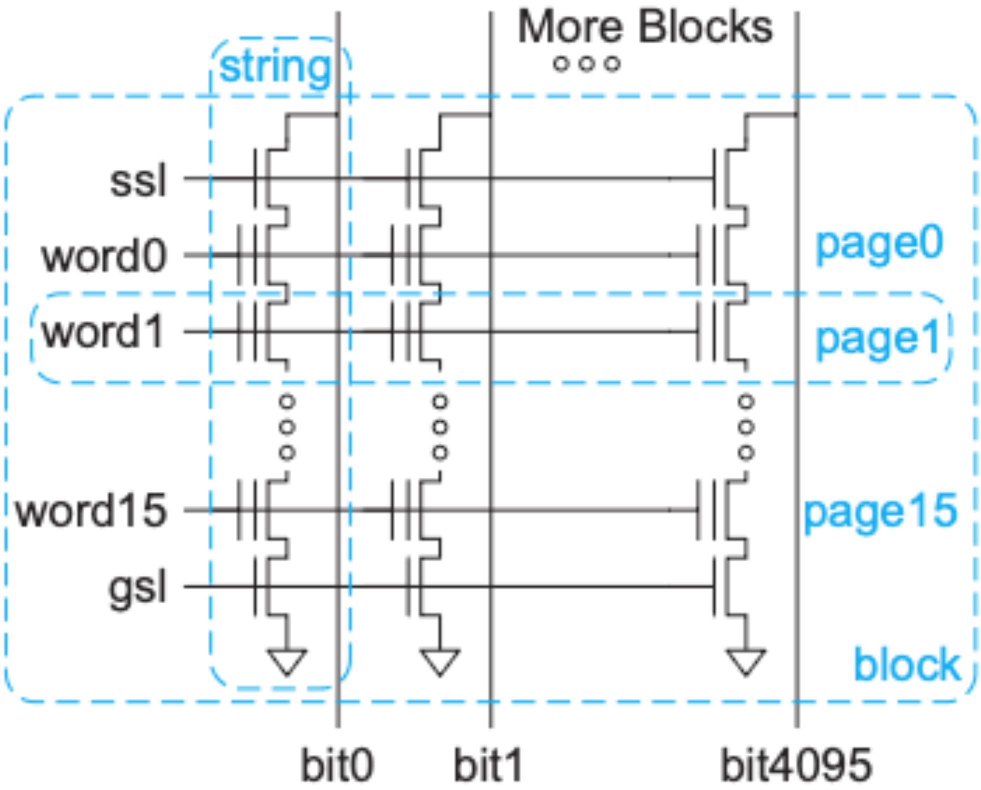
Flash Memory Cell



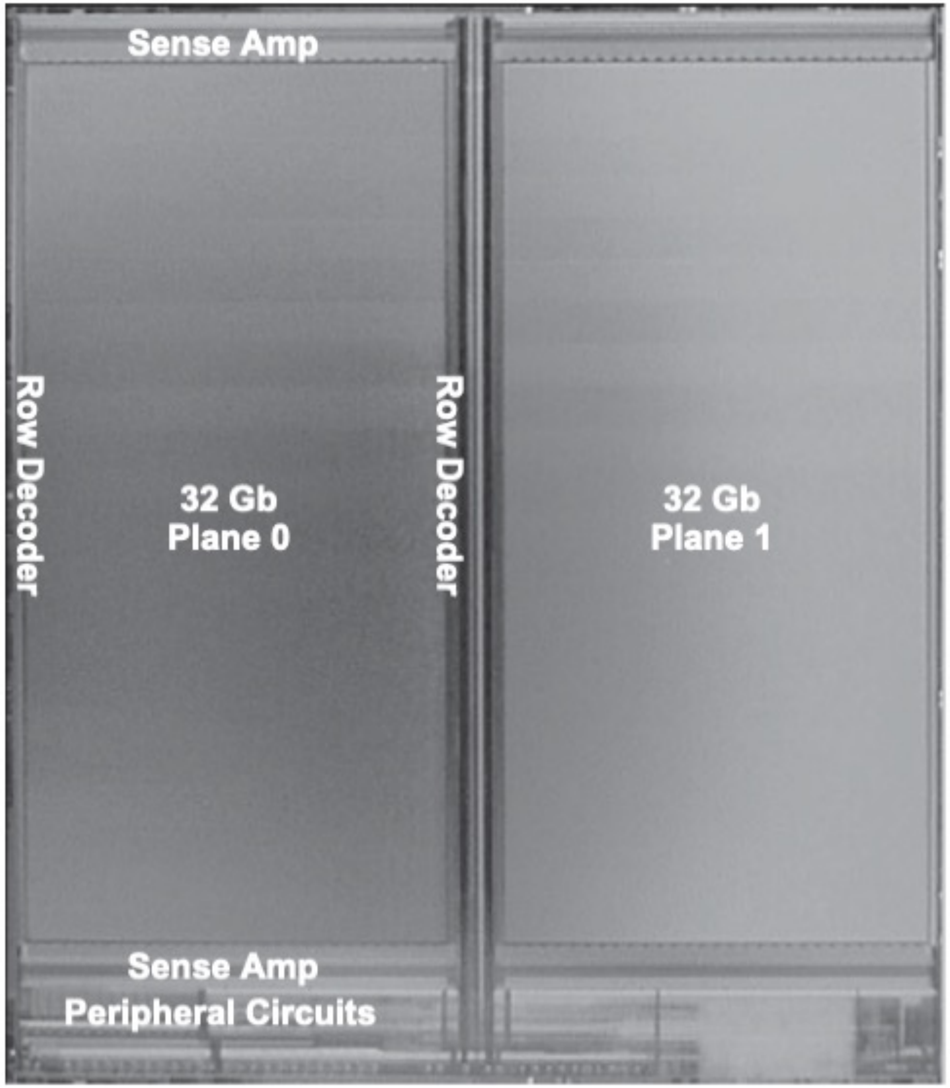


W&H p531

Flash Memory Array

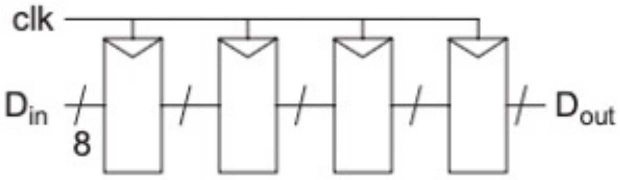


Each cell can store multibit value!



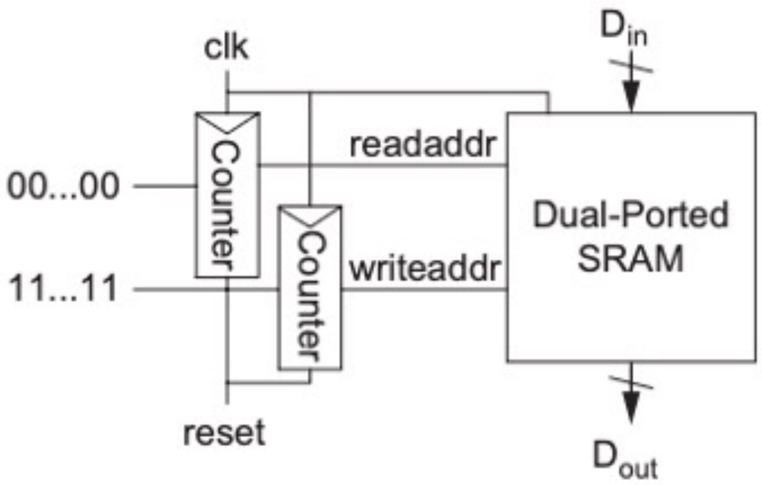
W&H p531

Shift Registers



(a)

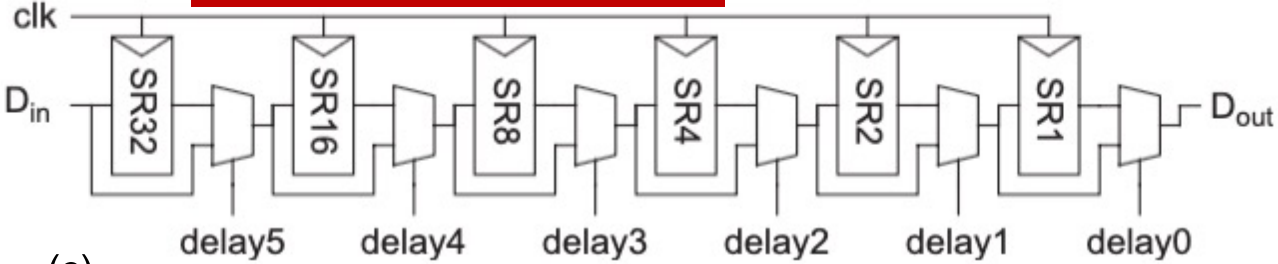
8-bit 4-stage SR



(b)

SR implemented with SRAM and counters

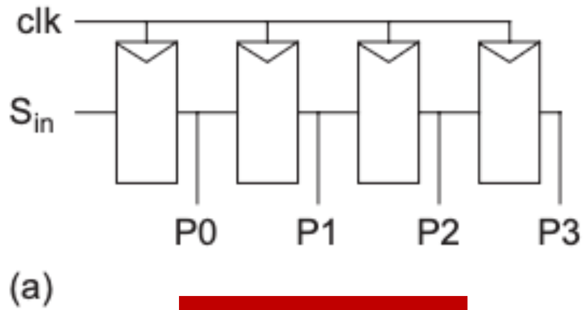
SR as tapped delay for DSP



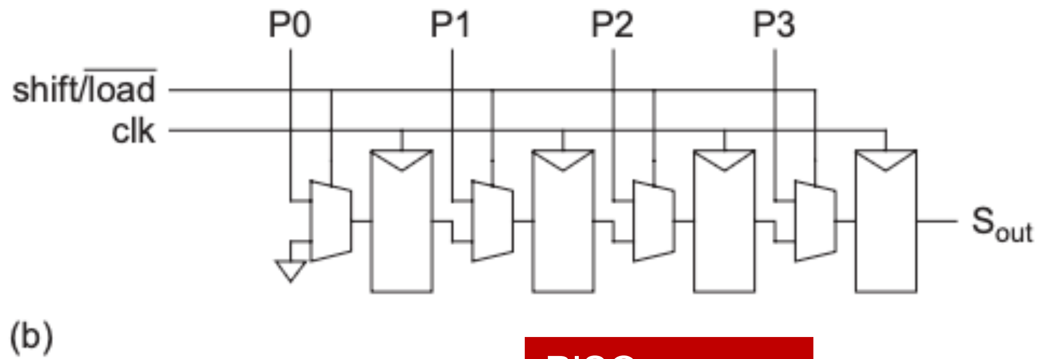
(c)

W&H p534

Delay Line and FIFO

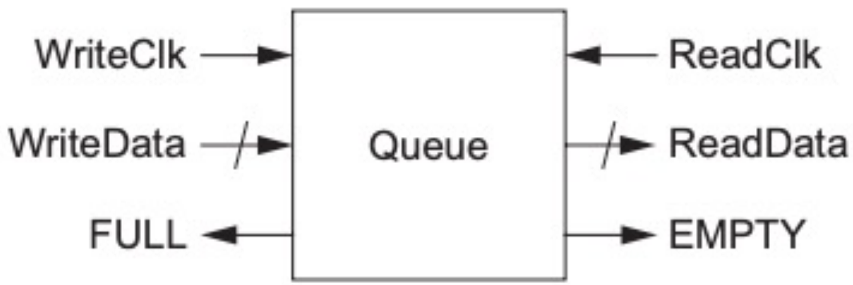


SIPO memory



PISO memory

FIFO buffer



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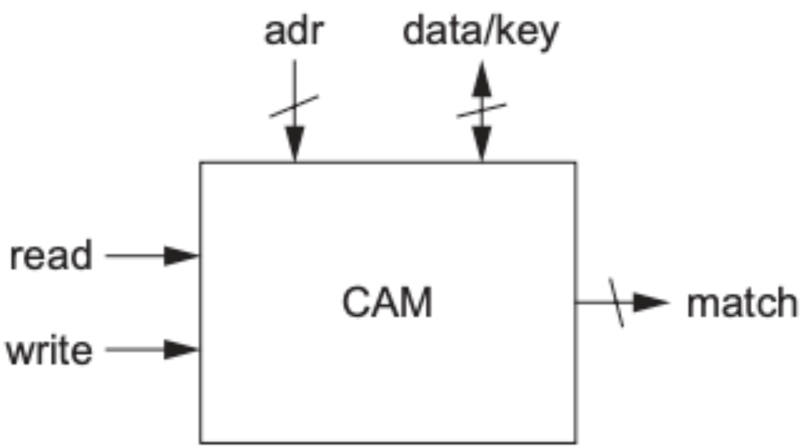
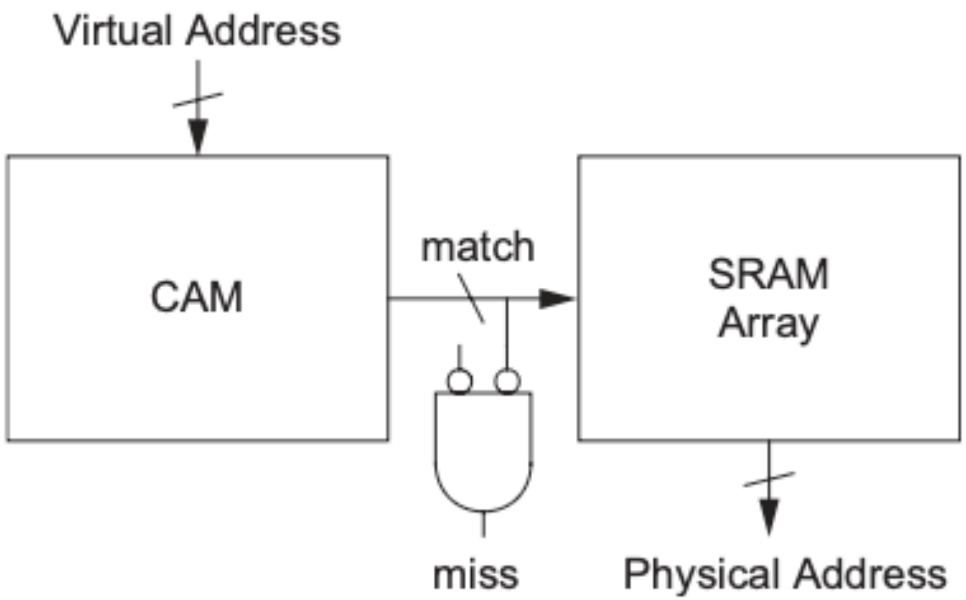
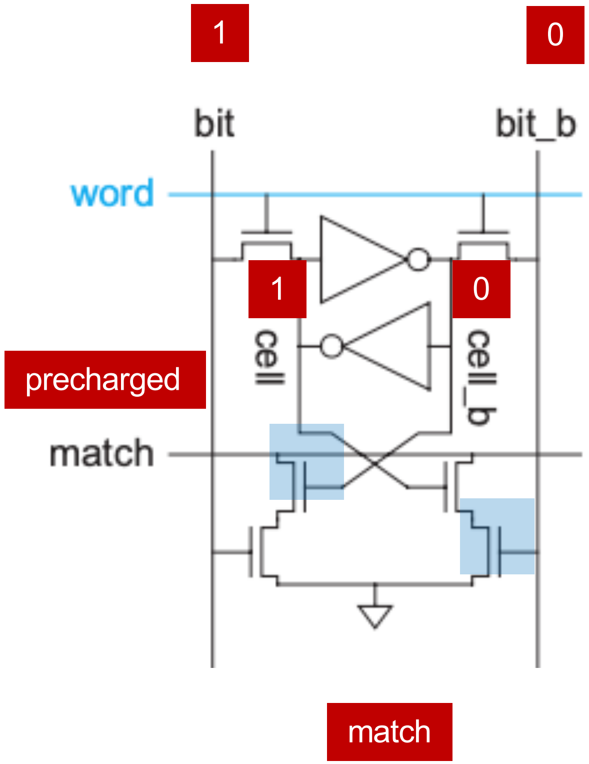
Content Addressable Memory (CAM)

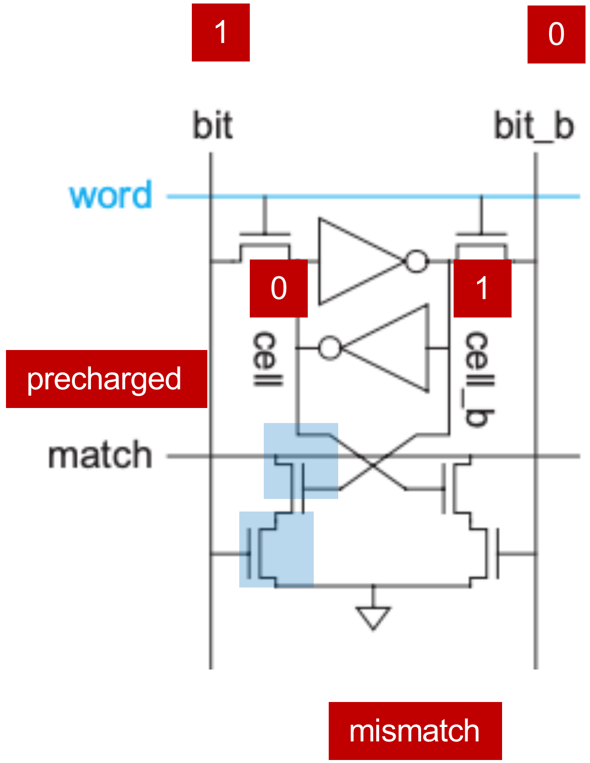




W&H p535

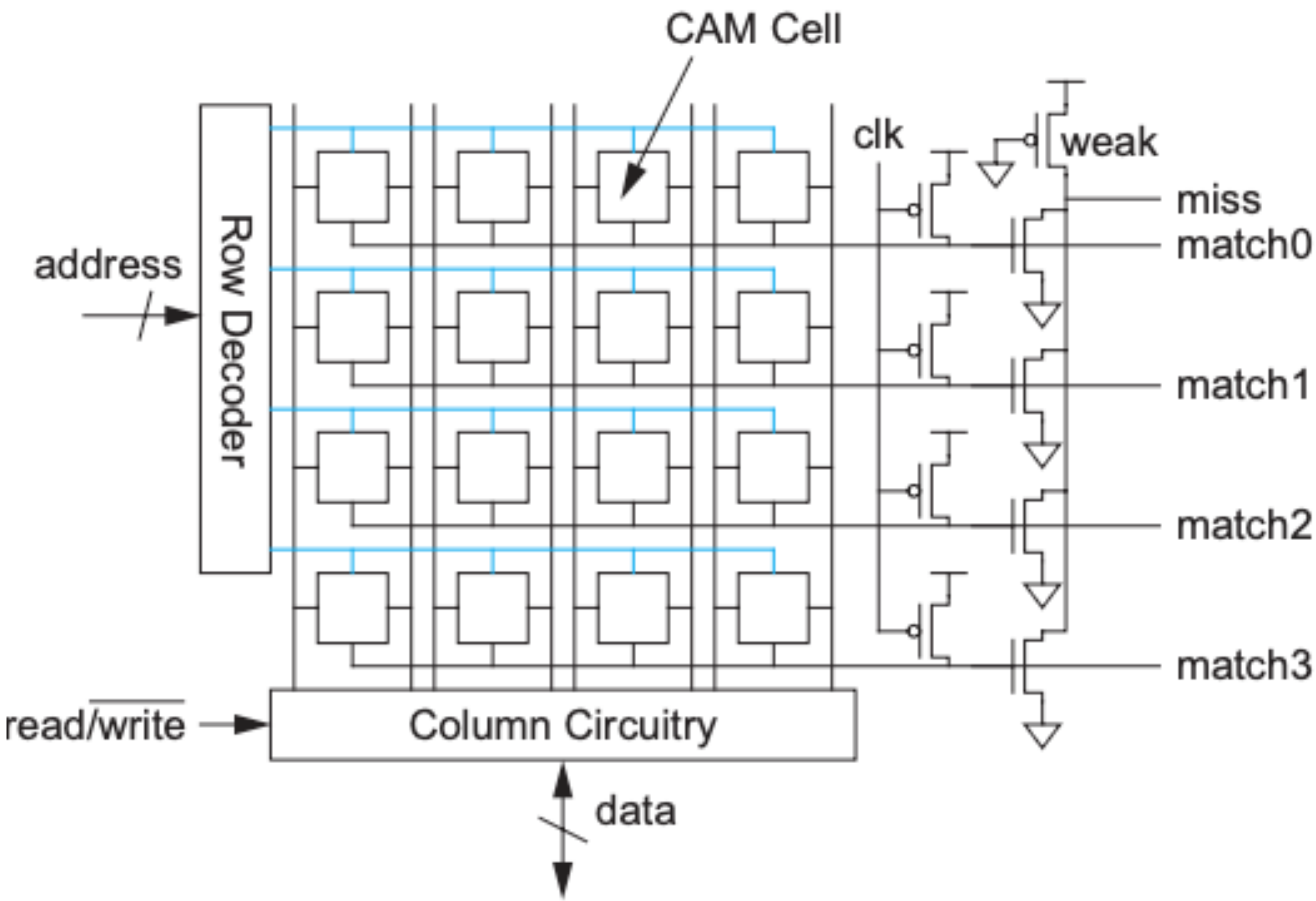
CAM Circuit (1)





W&H p535

CAM Circuit (2)



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